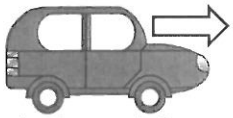
W.S.71. Force and movement.

Name .....

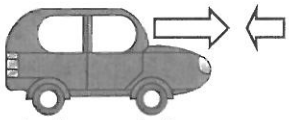
A FORCE is a PUSH or PULL. Force is measured in NEWTONS (N). Forces can speed up or slow down objects. The diagrams below show how different forces can affect the movement of a car.

1. Force from the engine makes the car begin to move.



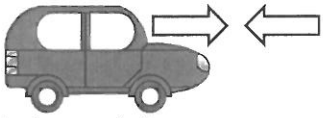
unbalanced force

2. As the car speeds up the force of air resistance gets bigger.



unbalanced force

3. The car reaches a steady speed when the two forces are equal.

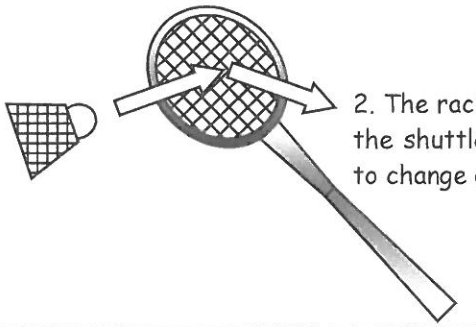


balanced force

When the force pushing against the car is the same size as the force from the engine the car stops accelerating and travels at a steady speed.

Forces can also make objects change direction. The diagram below shows this.

1. Shuttlecock moving in one direction hits the racket with a force.



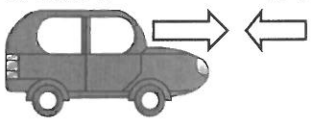
2. The racket gives a force to the shuttlecock and causes it to change direction.

The important rules from this are :

1. **Unbalanced forces change the speed and/or direction of moving objects.**
2. **Balanced forces produce no change in the movement of an object.**

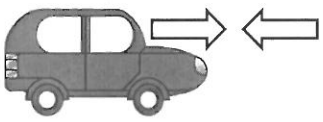
Exercise - Complete the sentences underneath each of the diagrams below.

Force from engine is 500N. Force of air resistance is 300N.



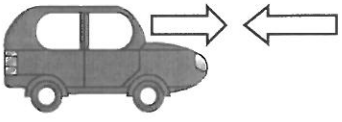
1) The car will _____

Force from engine is 500N. Force of air resistance is 500N.

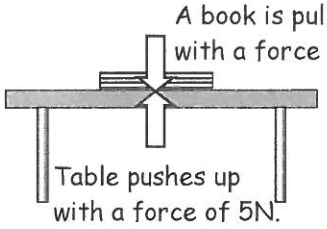


2) The car will _____

Force from engine is 500N. Force of air resistance is 700N



3) The car will _____



4) The book will not _____

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